AB

# PAPER CODE: BET-C610

## SEMESTER EXAMINATION (MAY-2025)

2263012012

## CLASS: B. TECH. SEMESTER: VI

## PAPER TITLE: EMBEDDED SYSTEMS

**Time: 3 Hours** **Max. Marks: 70**

**Note:** Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

| | | CO | BL |
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| | **SECTION-A (SHORT ANSWER TYPE QUESTIONS)**<br>**Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks. (5 x 6 = 30 Marks)** | | |
| Q-1 | In an RTOS-based system, Task 1 sends messages to Task 2 via a message queue. If Task 1 sends three messages to the queue, and Task 2 processes one of them, how many messages are left in the queue? | CO2 | L2 |
| Q-2 | Discuss Operating system Architecture approach. Which approach of OS designing is widely used these days is explained. | CO1 | L6 |
| Q-3 | Design an embedded system for a smart irrigation system. What hardware and software components would you use? | CO3 | L1 |
| Q-4 | Discuss addressing modes in 8051 microcontrollers. | CO1 | L6 |
| Q-5 | State the difference between FSM and FSMD models. | CO4 | L2 |
| Q-6 | Write an 8051-assembly language program to toggle Port 1 continuously with a 1-second delay between each toggle. | CO2 | L2 |
| Q-7 | Discuss the basic difference between timers and counters in the case of 8051 microcontrollers. | CO1 | L6 |
| Q-8 | Discuss the role of TMOD Register in case of Timer/Counter programming. | CO1 | L6 |
| Q-9 | Write a program to copy a block of 10 bytes of data from RAM locations starting at 35H to RAM locations starting at 60H. | CO3 | L5 |
| Q-10 | Explain the difference between hard real-time and soft real-time systems with examples. | CO2 | L3 |
| | **SECTION-B (LONG ANSWER TYPE QUESTIONS)**<br>**Instructions: Answer any four questions in detail. Each question carries 10 marks. (4 x 10 = Marks)** | **CO** | **BL** |
| Q-1 | Write a Program to find the GCD of two numbers, how can you optimize it. | CO3 | L2 |
| Q-2 | Define NRE Cost in short. The lifetime of a product is 58 weeks. If the product is delayed by 5 weeks, determine the percentage revenue loss? Determine the per product cost if NRE cost is Rs. 500000 and unit cost is Rs. 8000 and company produces 6000 product units of that product. | CO1 | L6 |



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| Q-3 | Write a program to interface 8 LED (Light Emitting Diode) with microcontroller 8051. | CO3 | L2 |
| Q-4 | Write a program in assembly language to generate a square wave of 2KHz frequency on Pin P1.5 of 8051 microcontrollers. | CO3 | L2 |
| Q-5 | Write a program to interface LCD (Liquid Crystal Display unit) with microcontroller 8051 | CO1 | L2 |
| Q-6 | Explain with a diagram of embedded system development life cycle. | CO3 | L4 |
| Q-7 | Discuss Message queues, and Mailboxes in detail. | CO1 | L6 |
| Q-8 | Explain the process of interfacing a 4x3 matrix keypad with the 8051. How is the keypress detected? | CO3 | L4 |

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